Audio Learning Lab — Sound Recording List

Sounds to record & upload for the Pro Audio Training Academy labs
Generated 2026-08-05

No lab currently plays a recorded clip — **all lab audio is synthesized** (math-generated teaching signals or the native tone/additive engine). Each sound below is a real recording that would replace or upgrade a synthesized stand-in, making the labs' waveforms, spectra and spectrograms show the real thing.

Recommended recording spec

Format	48 kHz / 24-bit **WAV**, mono, dry (no EQ, compression or reverb — the labs analyze them honestly). 48 kHz matches the audio engine.
Length	One-shots (kick, snare, cymbal hit, clap, bell): ~2–4 s including the natural decay. Sustained sources (organ, vocal, noise, hum, HVAC, drones): ~6–8 s.
Loop	**Sustained / continuous** sources should loop seamlessly — trim edits to zero-crossings. **One-shots** (transient hits) should NOT loop; they play once. Music mix: a few bars, loopable.
Level	Leave ~3 dB headroom — do not normalize to 0 dBFS. Keep a consistent peak across the set.
Naming	Lowercase, descriptive: e.g. kick.wav, snare.wav, organ_sustained.wav, bass_E1.wav, hum_60hz.wav, hvac.wav.

Drums & percussion

Sound	Type	Loop	Used in / purpose
Kick drum	One-shot	No	Visual Audio Analysis lab (waveform, spectrum); referenced in wave/calc examples
Snare	One-shot	No	Visual Audio Analysis lab (waveform / level)
Cymbal (+ long decay)	One-shot	No	Spectrum & spectrogram modules — bright, decaying content
Hi-hat	One-shot	No	Drum-kit set for level / transient reading
Clap	One-shot	No	Drum-kit set; clap-test / transient examples

Instruments

Sound	Type	Loop	Used in / purpose
Electric bass — open + fretted E / A / D / G	One-shot per note	Optional	Bass Guitar Physics lab (strings, frets, harmonics)
Electric guitar	One-shot / sustained	Optional	Visual Audio Analysis lab (waveform / spectrum)
Organ — sustained / drawbar	Sustained	Yes	Visual Audio Analysis lab (sustained tone with rich harmonics)
Bell / metallic tone	One-shot	No	FM Synthesis lab (the classic FM bell / inharmonic timbre)

Voice

Sound	Type	Loop	Used in / purpose
Sung vocal	Sustained notes	Per-note	Autotune lab (real retuning target instead of the synthesized voice)
Spoken speech	One-shot phrase	No	Visual Audio Analysis + spectrogram; future Speech lab

Reference

Sound	Type	Loop	Used in / purpose
Full-band music mix	Loopable phrase	Yes	Visual Audio Analysis lab — 'music mix' reference source (a few bars)

Noise Lab — real-world sources

These were explicitly called out as needing recorded assets. Most are continuous — record clean, loopable takes.

Sound	Type	Loop	Used in / purpose
Mains hum (record both 50 Hz and 60 Hz)	Sustained	Yes	Noise lab real-world sources; hum examples in meter labs
Electrical buzz	Sustained	Yes	Noise lab
Ground-loop hum	Sustained	Yes	Noise lab
HVAC / air handler	Sustained	Yes	Noise lab
Traffic	Sustained	Yes	Noise lab
Wind	Sustained	Yes	Noise lab
RF interference	Sustained / intermittent	Yes	Noise lab
Crackle	Texture	Yes	Noise lab
Static	Sustained	Yes	Noise lab
Grey noise	Sustained	Yes	Noise lab (can also be synthesized)
Birdsong	One-shot / loop	Optional	Spectrogram module — tonal sweeps
Whistle	Sustained	Yes	Spectrogram module — pure tone glide
Feedback tone	Sustained	Yes	Spectrum module — rising feedback example

---

Note: white, pink, brown, blue and violet noise are already generated natively and do not need recording. Waveform primitives (sine, square, triangle, saw, pulse) are synthesized too.